FINTECH 545 – Assignment 2

Nils Viehweger

Problem 1

a) Building the correlation matrix pairwise can come back as non-PSD. This is because every entry in the matrix comes from a different subsample (each pair uses every row where both legs of the pair are observed, regardless of whether the other assets are observed or not). There is no single dataset behind the matrix, so entries can be mutually inconsistent. Using the complete case, the matrix is guaranteed to be PSD, as it will stem from one dataset. The problem with that is that there are only 81 observations (out of 250 days) where all assets trade, so we would lose a lot of information.

b) The index being an approximate (due to tracking error) linear combination of the four assets implies that the smallest eigenvalue of the true correlation matrix will be approximately zero. Given that, the matrix already sits at the PSD boundary. Noise in individual entries can then push an estimate to be negative. This is most likely to occur for asset D where noise is largest as that asset has only 96 observations.

c) I expect the correlation between assets C and D to be the least reliable, as that pair only has 83 joint observations. Any D-related entries might not be the most reliable as, again, that asset only has 96 observations. I expect the correlation of assets A and B (and each with the index) to have the most reliable correlation out of the bunch as there are 242 joint observations.

d)

(d) Eigenvalues, ascending:

	1	2	3	4	5
Complete case	0.001506	0.548345	0.693320	0.870441	2.886388
Pairwise	**-0.011072**	0.471422	0.528367	0.857381	3.153902

Cholesky: complete case **succeeded**; pairwise **failed** ("Matrix is not positive definite").

The Cholesky factorization succeeds for in the complete case, as the correlation matrix is PSD, while it fails in the pairwise case due to a negative eigenvalue.

e) The variance of the portfolio that is long one unit of IDX and short the weighted basket comes in negative (-1.799370e-06) using pairwise correlations and the full

history standard deviation of each series.

```
(e) Covariance from pairwise correlations and full-history standard deviations
================================================================================
Full-history standard deviations: A=0.016502  B=0.011919  C=0.018324  D=0.022829  IDX=0.012281

Covariance matrix (pairwise correlations x full-history SDs)
             A         B         C         D       IDX
A    0.000272  0.000092  0.000130  0.000165  0.000176
B    0.000092  0.000142  0.000101  0.000075  0.000107
C    0.000130  0.000101  0.000336  0.000077  0.000161
D    0.000165  0.000075  0.000077  0.000521  0.000170
IDX  0.000176  0.000107  0.000161  0.000170  0.000151

Tracking portfolio: long 1 IDX, short 0.40 A, 0.30 B, 0.20 C, 0.10 D
Tracking portfolio variance (pairwise matrix) = -1.799370e-06
  -> NEGATIVE: a variance below zero, which no real portfolio can have.
Tracking portfolio variance (complete-case correlations, same SDs) = 1.097545e-06
```

f) The repair fixed the failure from e). Tracking variance is now >0.

```
Summary (f):
                    method  smallest_eigenvalue  frobenius_from_pairwise  tracking_variance
Rebonato-Jackel (near_psd)         1.468910e-16             1.618210e-02       6.807298e-07
                    Higham        -9.912135e-10             1.552000e-02       6.787647e-07
```

g) The tracking portfolio variance in e) corresponds to an eigenvalue and the weight vector to the corresponding eigenvector. By the definition of positive semi-definiteness, the covariance matrix is non-PSD, as an eigenvalue (interpreted here the tracking portfolio variance) is negative. This means that the pairwise correlation matrix assigns a negative variance to a real portfolio, meaning that the matrix cannot be a valid covariance matrix. The fixes in f) repair the covariance matrix by making all eigenvalues >= 0.

h) Higham moved correlation entries that are paired with IDX the most. In part c), I predicted that correlations involving asset D would be the least reliable, but in actuality, they were some of the least changed ones. This is due to the fact that the change after using Higham is not a measure of whether a correlation is reliable. The Higham repair has no notion of how many days sit behind an entry. Higham looks for the nearest valid matrix that gets rid of the negative eigenvalue. It is aiming for smallest total change. It uses the eigenvector of the negative eigenvalue to decide where to make changes.

```
Higham: pairs ranked by size of change, with the days each pair is jointly observed
 pair    change  joint_days
A-IDX -0.007045         242
C-IDX -0.004765         228
B-IDX -0.004180         242
D-IDX -0.003727          96
  A-C  0.002277         221
  A-B  0.001997         242
  A-D  0.001781          93
  B-C  0.001351         221
  C-D  0.001205          83
  B-D  0.001057          93
```

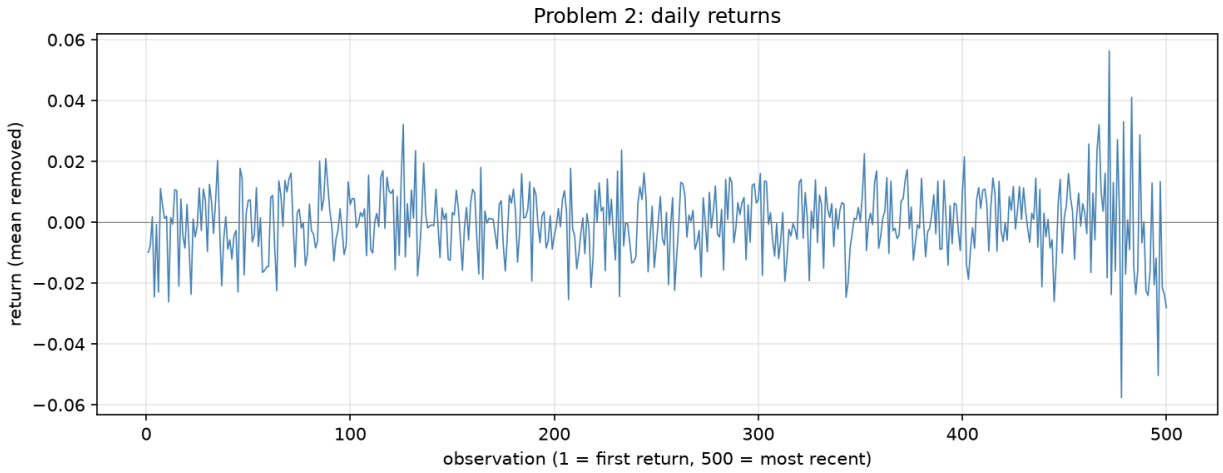
i) The size of the gap between complete case versus pairwise is much larger than the repair sizes for either choice of repair. The two repairs barely differ from each other. Therefore, the decision of estimator matters more than the choice of repair.

```
(i) Size of the estimator gap vs size of the repairs (Frobenius distances)
================================================================================
complete case vs pairwise        = 0.423935
Rebonato-Jackel (near_psd)   repair size = 0.016182
Higham                       repair size = 0.015520
```

Problem 2

a) 4 → 5 → 1 → 2 → 3

Recent volatility is much higher than the historical mean. VaR estimates based on distributions that with an exponentially weighted standard deviation will give more weight to the recent period and therefore have a higher volatility estimate and therefore larger VaR estimate. The Student's t is below the equal weighted normal as the 95% level, the Normal is wider than the t distribution, as found in Assignment 1. The historical sample is also equal weight, so it fits into the group of equal weighted options here. I have placed it in between the Normal and t distribution based on what I've seen from Assignment 1, but since its essentially just the 25th worst day of 500, there is real sample noise that makes it hard to say where it would end. The normal with exponentially weighted standard deviation at lambda of 0.97 would give a lower VaR estimate than the one at lambda of 0.94, as the latter puts more weight on recent observations (decays faster), and the recent observations are the ones with heightened volatility.



b) Looking at the plot, it seems like just over 30 days are different than the rest. At lambda = 0.94, ~84% of the weight is placed on the most recent 30 days versus ~60% of weight using lambda = 0.97.

```
(b) Effective sample size and half life
========================================
  lambda      n_eff   half life (days)
    0.94      32.33              11.20
    0.97      65.67              22.76

(b) Share of total weight on the most recent k days
===================================================
     k    lambda=0.94    lambda=0.97   equal weight
     5         0.2661         0.1413         0.0100
    10         0.4614         0.2626         0.0200
    20         0.7099         0.4562         0.0400
    30         0.8437         0.5990         0.0600
    40         0.9158         0.7043         0.0800
    50         0.9547         0.7819         0.1000
    75         0.9903         0.8982         0.1500
   100         0.9979         0.9524         0.2000
```

c) A single normal regime implies skewness and excess kurtosis of 0. This sample has a skewness of -0.156 and an excess kurtosis of 2.166. Skewness is within about 1.4 standard errors of 0 while excess kurtosis is within about 9.9 standard errors. The skewness could plausibly be produced by a single regime of normally distributed returns, but the observed excess kurtosis is too extreme to plausibly have been produced by a Normal.

```
(c) First four moments of the full sample (mean removed)
=========================================================
mean            = -2.220446e-19
variance        =  1.418981e-04
std deviation   =  0.011912
skewness        = -0.155587
excess kurtosis =  2.132367
```

d)

```
(d) One-day 5% VaR on $1,000,000, five ways
==========================================
estimate                            SD of the model       VaR ($)   VaR (% of position)
1. Normal, equally weighted SD             0.011912        19,594              1.9594%
2. Normal, EW SD, lambda = 0.97            0.020709        34,064              3.4064%
3. Normal, EW SD, lambda = 0.94            0.022074        36,308              3.6308%
4. Student's t (MLE)                     0.011815 *        19,057              1.9057%
5. Historical simulation                        --         20,574              2.0574%
* implied SD of the fitted t; its VaR is computed from the fitted t's parameters below, not from this SD.
-- historical simulation assumes no distribution, so there is no model SD.

Smallest to largest: 4 < 1 < 5 < 2 < 3

Fitted Student's t parameters (full sample, mean-removed returns):
  location mu                  = -1.9785e-19   (effectively 0)
  scale sigma                  = 0.010306
  degrees of freedom nu        = 8.3606
  implied SD = sigma*sqrt(nu/(nu-2)) = 0.011815
```

e) 5 and 1 came in flipped from how I had predicted them. The VaR estimates for both are fairly close with the equal weight Normal coming in at ~$19,600 and the Historical Simulation at ~$20,600. I had stated that there really is no way of predicting where exactly the historical VaR would land due to sampling noise. I would expect the true distribution of historical returns to have a fatter left tail than the normal though, but predicting at exactly the 95% level is not manageable at this sample size. Other than that, things landed where expected.

f) When split into two regimes, the excess kurtosis is close to 0 in both. Looking at the full sample is when the excess kurtosis comes in. Basically, the returns follow a Normal process (based on kurtosis) in each regime, and the non-Normal kurtosis observed over the full sample is assuming one Normal distribution for a process that really involves two Normals with different volatilities. The fitted t is actually describing a summary of the calm and the more volatile regimes. Its scale is close to the calm periods SD, while its implied SD is close to the full-sample SD. The heavy tails (nu = 8.36) absorb the period with heightened volatility.

```
(f) Volatility inside each regime
================================
Split: first 470 returns (calm) | last 30 returns (recent)

                       n          SD    skewness  excess kurt.
calm                 470    0.010379     -0.1399       -0.1178
recent                30    0.025593      0.5251        0.1268
full sample          500    0.011912     -0.1556        2.1324

ratio of SDs (recent / calm) = 2.466
```

g) The difference between VaRs at the two lambda levels is within estimation noise.

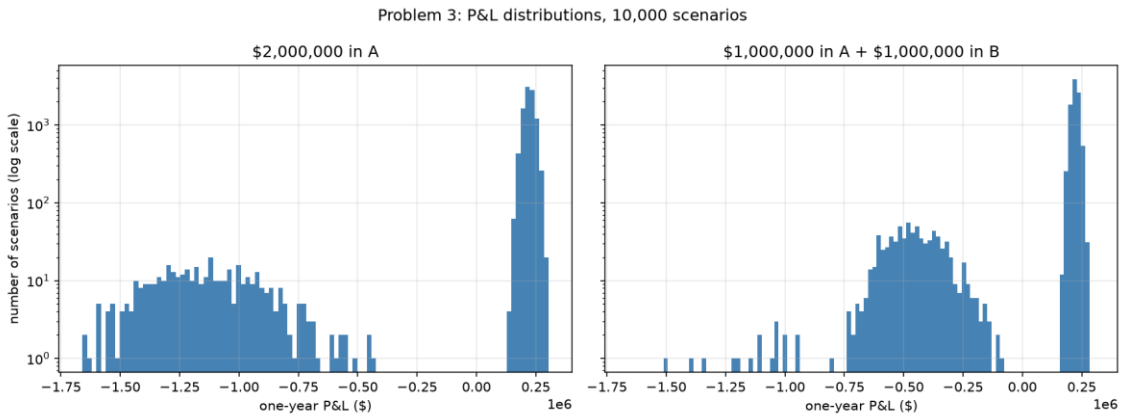
```
(g) Is the gap between the two EW VaRs bigger than the estimation noise?
========================================================================
  lambda    n_eff   SE of vol (relative)          SD    SE of SD
    0.97    65.67                  8.73%    0.020709    0.001807
    0.94    32.33                 12.44%    0.022074    0.002745

VaR at lambda = 0.97: $34,064
VaR at lambda = 0.94: $36,308
gap = $2,244  (6.59% of the lambda = 0.97 VaR)
```

The difference between which lambda to choose is less impactful than deciding whether to use an equal or exponentially weighted standard deviation.
Opinion: I would use the lower lambda as a trading limit, and the larger lambda for a capital number. Disregarding strategy specific details, I would want my trading limit to respond faster and weight recent observations even more heavily, even though that comes with nosier estimates. A capital number should be more stable and have a longer memory in my opinion. It needs to be defensible and should not be hypersensitive, as it has broader implication than a trading limit would have.

Problem 3

a)




```
(a) First four moments of each bond's return
============================================
                        A         B
mean             0.083709  0.085011
SD               0.135988  0.132102
skewness        -4.822523 -4.914778
excess kurtosis 22.081997 23.074367
min             -0.829748 -0.855977
max              0.151810  0.154444
```

A normal VaR on these positions is not reliable; it assumes a continuous and

symmetric distribution. The simulations show that outcomes are differently distributed based on whether there is a default or not with no overlap of the two distributions. The skewness and excess kurtosis numbers immediately show that modeling this process with a Normal is not a good idea.

b) We saw in class that diversification can raise VaR. The diversified choice should have a larger VaR because there is a 7.82% chance of at least one default occurring, which sits above the five percent level. Only investing in one of the two leaves the chance of observing a default around 4% (in this simulation) which is below the 5% level. So a 95% VaR would not reflect it. ES on the other hand will reflect the diversification benefit, as it shows the average loss in the worst 5% of scenarios rather than the location of the specific quantile. There's only 17 out of 10,000 scenarios in which both bonds defaulted, so ES will be a lot better for the diversified portfolio compared to the non-diversified one.

```
(b) Scenarios with a loss of more than 20%
==========================================
A loses more than 20%          406  (4.06% of 10,000)
B loses more than 20%          393  (3.93% of 10,000)
at least one of the two        782  (7.82% of 10,000)
both                            17  (0.17% of 10,000)
```

c)

d)

```
(c) Historical and (d) Normal VaR / ES at 5%, in dollars
=======================================================
VaR and ES are positive for a loss; a negative VaR means the 5% mark is itself a gain.
Normal VaR uses each P&L's own sample mean and SD (shown); hist ES is the average of the worst 5%.

                           hist VaR       hist ES     Normal VaR       P&L mean         P&L SD
position
$1M in A                     -84,531       443,628        139,971         83,709        135,988
$1M in B                     -86,011       419,587        132,277         85,011        132,102
$2M in A (concentrated)     -169,061       887,257        279,941        167,418        271,975
$1M in each (diversified)    414,767       541,997        143,466        168,721        189,796
```

e)

```
(e) The same at 1%, in dollars
==============================
                           hist VaR       hist ES     Normal VaR       P&L mean         P&L SD
position
$1M in A                     650,545       709,548        232,645         83,709        135,988
$1M in B                     641,248       700,805        222,303         85,011        132,102
$2M in A (concentrated)    1,301,090     1,419,096        465,291        167,418        271,975
$1M in each (diversified)    598,237       716,408        272,811        168,721        189,796
```

f) VaR can fail subadditivity when diversification pushes the total probability of loss across the confidence threshold because VaR ignores the size of losses beyond the

quantile. ES averages the tail, so it does not fail subadditivity.

```
(f) Subadditivity check: risk(A + B) <= risk(A) + risk(B)?
============================================================

At 5% (historical):
                $1M A         $1M B           sum      $1M each   combined - sum
VaR           -84,531       -86,011      -170,542       414,767          585,309
        subadditivity FAILS (combined risk exceeds the sum)
ES            443,628       419,587       863,216       541,997         -321,218
        subadditivity holds

At 1% (historical), for part (h):
                $1M A         $1M B           sum      $1M each   combined - sum
VaR           650,545       641,248     1,291,793       598,237         -693,556
        subadditivity holds
ES            709,548       700,805     1,410,353       716,408         -693,945
        subadditivity holds
```

g) Normal VaR favors diversified. That is the right choice, but for the wrong reason. Normal VaR is about half for diversified versus concentrated. Normal VaR selects diversified due to less variance which it reports as less risk. In reality, diversifying halves the size of each default loss but roughly doubles the chance that something defaults (~4% vs 7.8%). Normal VaR has no notion of these risk dynamics or default clustering.

h) Results change at the 1% level because each bond individually defaults in around 4% of scenarios. The 5% level therefore happens to be a non-defaulting scenario, while the 1% level is a default scenario. VaR's failure shows up when there a discrete, rare events cluster below the chosen threshold, as seen in the bond example here. Expected shortfall does not have this failure mode, as it averages everything beyond the threshold.

Problem 4

a)

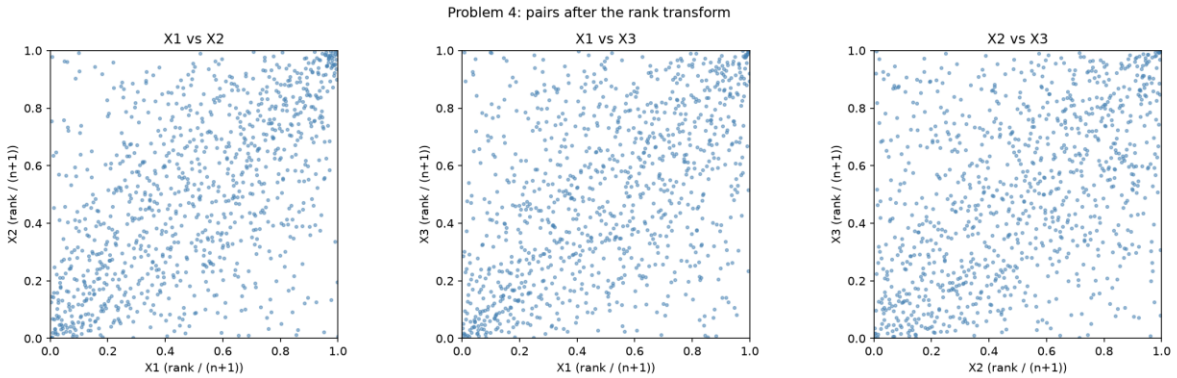
```
(a) First four moments of each series
====================================
                       X1        X2        X3
mean             0.000257 -0.000252  0.000890
SD               0.011894  0.016168  0.010019
skewness        -0.034174 -0.614117  0.073955
excess kurtosis -0.122542  7.083388  2.853745
min             -0.040000 -0.142914 -0.061244
max              0.033149  0.065627  0.053039

(a) Sensitivity to a single observation
=======================================
Moments recomputed after dropping the one observation furthest from the mean.
Use this if one of the numbers above looks out of place.

series  dropped value  obs #  excess kurt (all)  excess kurt (dropped)  skew (all)  skew (dropped)
    X1      -0.040000    952          -0.122542              -0.192699   -0.034174       -0.005487
    X2      -0.142914    952           7.083388               1.710093   -0.614117        0.055783
    X3      -0.061244    952           2.853745               1.716058    0.073955        0.312681
```

I would fit a normal distribution to X1. Based on only first four moments, I would likely fit a student's t distribution with different degrees of freedom for X2 and X3. After removing the most extreme observation, the excess kurtosis of X2 and X3 drop to ~1.71. X2's skewness flips from -0.61 to 0.05, and the student's t choice does not look as attractive anymore. After removing the most extreme observation, both X2 and X3 look somewhat closer to a normal distribution than before. The excess kurtosis of ~1.71 is still around 11 standard errors from zero. Therefore, I would keep the choice of a student's t for X2 and X3 but would go with a larger nu.

b)



```
(b) Days both series sit in their own tail
========================================
Tail size: 25 days (2.5% of 1000) per series.
Expected joint-tail days under independence: 1000 x 0.025 x 0.025 = 0.625

pair         both worst 2.5%    both best 2.5%
X1-X2                      6                 8
X1-X3                      7                12
X2-X3                      6                 9
```

Under independence, I would expect 0.625 days in for both series of a pair to sit in their own worst/best 2.5%. The plots do suggest that the joint tails are more connected than the middle of the distribution, as clear clusters are visible near the origin and near (1,1). The counts also suggest that, as they are far above the expected 0.625 days across pairs and across worst and best buckets. Given the clustering in the corners of the plot, I would expect the t Copula to win. It will depend on how the numbers play out though. The t copula models tail dependence (which is 0 for the gaussian copula) but it comes at the cost of estimating one extra parameter (nu) which AICc and BIC penalize. So if the benefit of modeling positive tail dependence outweighs the extra estimated parameter penalty, which I think it will, the t copula wins. Qualitatively, the behavior of the moments when removing that extreme observation also indicate strong tail dependence as that observation was extreme across assets as well.

c) X1 → Normal

X2→ t with 5.27 degrees of freedom

X3 → t with 6.09 degrees of freedom

```
(c) Normal vs Student's t for each margin, by AICc (lower is better)
======================================================================
        Normal mu  Normal sigma  Normal AICc      t mu   t sigma       t nu       t AICc  AICc gap (Normal - t)
series
X1       0.000257      0.011894 -6022.536012  0.000257  0.011888  8.356e+13 -6020.524453              -2.011560
X2      -0.000252      0.016168 -5408.493536 -0.000176  0.012544   5.271477 -5518.184759             109.691223
X3       0.000890      0.010019 -6365.690739  0.000746  0.008181   6.094881 -6428.015517              62.324778

Chosen margin: X1 = Normal, X2 = t, X3 = t
(nu above 1e6 means the t fit collapsed to a Normal: same likelihood, one more parameter.)
```

d)

```
(d) Copulas fitted to the uniforms from the chosen margins
============================================================
Correlation matrix from Kendall's tau (rho = sin(pi*tau/2)), shared by both copulas:
          X1        X2        X3
X1  1.000000  0.583263  0.502851
X2  0.583263  1.000000  0.464383
X3  0.502851  0.464383  1.000000

          log likelihood  nu-hat  k      AICc       BIC
Gaussian        350.4536      --  0 -700.9072 -700.9072
t               451.9988  3.6585  1 -901.9936 -897.0898

t minus Gaussian:  log likelihood +101.5452   AICc -201.0864   BIC -196.1826   (negative = t preferred)
```

e)

```
(e) Portfolio risk, $1,000,000 in each of X1, X2, X3 (dollars, positive = loss)
================================================================================
                                    VaR 5%       ES 5%       VaR 1%       ES 1%
Gaussian copula (100,000 sims)      49,546      65,983      75,481      92,945
t copula (100,000 sims)             49,032      68,451      79,235     101,768
Historical (1,000 days)             47,939      65,699      70,092     102,424

Simulation noise: the same risk numbers over 10 different seeds (mean and SD):
                    VaR 5%       ES 5%       VaR 1%       ES 1%
Gaussian: mean      49,515      66,195      75,840      93,691
Gaussian: SD           241         313         436         805
t: mean             48,940      68,062      78,934     100,648
t: SD                  279         516         751       1,132
```

f)

```
(f) Joint tail days over 1,000 days: each copula's, next to the data's
======================================================================
pair              both worst 2.5%                       both best 2.5%
             data      Gaussian            t      data      Gaussian            t
X1-X2           6          5.96         9.81         8          5.78         9.17
X1-X3           7          4.49         8.07        12          4.58         7.64
X2-X3           6          4.17         7.42         9          4.18         7.42

(independence would give 0.625 days)
```

g) The t copula (nu-hat = 3.66) wins across different criteria:

a. Log likelihood: 451.999 (t) against 350.454 (Gaussian), a gain of 101.5.
b. AICc: -901.99 against -700.91, so the t is better by 201.1.
c. BIC: -897.09 against -700.91, so the t is better by 196.2.

On the joint tail counts: the gaussian copula implies roughly 4 to 6 days per pair in each tail versus the t implying 7 to 10. The data sits above the Gaussian in all six cells. The t sits above the data four out of six times. The Gaussian understates the joint extremes in the data, while the t overstates it (usually by a smaller margin as

seen in the screenshot in f). Given that we are modeling risk, the t copula's more conservative estimates are preferable.

h) It moves the 1% ES the most (+9.5% for t). The 5% VaR barely moved (-1.0%) because the difference between gaussian vs t copula affects tail dependence. The effect becomes more noticeable the further out on the tail you look. VaR at 5% is a quantile where the two copulas have similar joint behavior.

i)

```
(i) Each day's contribution to (log likelihood of t copula - log likelihood of Gaussian copula)
================================================================================================
check: sum of daily gaps = 101.5452  vs  total gap 101.5452

Days with no series in an outer 5% on either side: 793 of 1000
  their total contribution:           +50.5182   (49.7% of the total)
Days with at least one series in an outer 5%: 207 of 1000
  their total contribution:           +51.0270   (50.3% of the total)

Five days with the largest contribution (observation number, contribution, uniforms):
  obs  642   +10.2890   u = 0.0076  0.9789  0.9944
  obs  481   +10.2459   u = 0.9972  0.0037  0.9279
  obs   87    +6.0943   u = 0.0947  0.9938  0.9937
  obs  384    +4.9489   u = 0.4156  0.0104  0.9987
  obs  649    +4.2214   u = 0.0553  0.9599  0.0126

Days where the t copula is the better fit (contribution > 0): 579 of 1000
```

49.7% of the total comes from days in which no series sits in its outer 5% on either side. That tells us that the information criterion is not only rewarding joint crashes. The five days with largest contributions actually have series moving in opposite directions. The t's common shock scales all series up together, meaning it assigns far more probability than the gaussian to days when series are large in magnitude at once no matter whether up or down. The gaussian with positive correlations treats opposite extremes as nearly impossible, so it performs poorly on such days. AICc and BIC score how well the whole joint density fits the 1,000 days minus a penalty. The t outperforms the gaussian on extreme days no matter the direction, but also, by a smaller per observation margin, on ordinary days.

j) X1–X2 has a t-copula tail dependence of 0.322 and a Gaussian one of 0.

```
(j) Lower tail dependence coefficient
=====================================
pair       correlation    t copula    Gaussian
X1-X2           0.5833      0.3220      0.0000
X1-X3           0.5029      0.2733      0.0000
X2-X3           0.4644      0.2525      0.0000

Most correlated pair: X1-X2 (correlation 0.5833), fitted nu = 3.6585
```

Problem 5

a) Variance formula for a two-stock portfolio:
$\text{var}(w_A \cdot r_A + w_B \cdot r_B) = w_A^2 \cdot \text{var}(r_A) + w_B^2 \cdot \text{var}(r_B) + 2 \cdot w_A \cdot w_B \cdot \text{cov}(r_A, r_B)$
Assuming a shared industry factor that is not captured by the model makes the residual covariance positive, affecting the cross term (the one including the covariance). For P1, that would understate the portfolio variance. The product of weights in the cross term for P1 is positive, adding variance → therefore the model understates variance and therefore also understates VaR. For P2, the weight product is negative, so the cross term subtracts variance. Setting it to zero (aka assuming independence) would remove the subtraction, and therefore would overstate P2's variance and VaR. The shortcut would miss the benefit from the offset of the shared industry factor in the long-short portfolio.

b)

```
(b) Regressions of each stock on the market: r_i = alpha + beta * r_MKT + eps
================================================================================
500 daily arithmetic returns; residual variance divides by T - 2 = 498

                alpha        beta   residual SD
A            0.000205    1.035278      0.012901
B            0.000315    0.935417      0.011550

Residual covariance matrix:
             A           B
A  1.6644e-04  1.0405e-04
B  1.0405e-04  1.3340e-04

Correlation between the residuals of A and B = 0.698307
Market return variance = 1.1136e-04   (SD 0.010553)
```

c)

```
(c) Simulated VaR at 5%, dollars (100,000 days)
================================================
                    full residual cov off-diagonal zero shortcut effect
portfolio
P1: long A, long B             50,411            44,570          -11.6%
P2: long A, short B            15,951            28,731          +80.1%

Same models in closed form (no simulation noise):
                    full residual cov off-diagonal zero shortcut effect
portfolio
P1: long A, long B             50,442            44,512          -11.8%
P2: long A, short B            15,849            28,535          +80.0%

Simulation noise: SD of the simulated VaR over 10 seeds
  full residual cov   P1: long A, long B: $278   P2: long A, short B: $81
  off-diagonal zero   P1: long A, long B: $181   P2: long A, short B: $150
```

d)

```
(d) Delta normal VaR at 5% from the sample covariance of the two stock returns, dollars
=======================================================================================
Sample covariance matrix of (A, B) returns:
            A           B
A  2.8546e-04  2.1169e-04
B  2.1169e-04  2.3058e-04

P1: long A, long B      $      50,415
P2: long A, short B     $      15,834
```

e) The results show that the shortcut did indeed move in the direction I had predicted. The L/S portfolio is more exposed. The shortcut omits the residual covariance term. While it is the same size in both portfolios, it is positive in P1 and negative in P2. In P1, it is about 22% of a large variance. In the L/S P2, the market factor mostly cancels as the stocks have similar betas. The correlated residuals also offset each other, so the true variance is small. The omitted term is about 224% of the true variance. This is why there is such a large discrepancy in how much the shortcut affects the VaR estimates for the two portfolios.

f)

```
(e)/(f) Side by side, dollars
============================
                      delta normal  full resid (closed form)  full resid (simulated)  off-diag zero (simulated)
portfolio
P1: long A, long B          50,415                    50,442                  50,411                     44,570
P2: long A, short B         15,834                    15,849                  15,951                     28,731

Why the full-residual and delta normal figures should agree: the model's covariance of (A, B)
is beta beta' var(MKT) + residual covariance. Compare it with the sample covariance:
           A          B
A  3.336e-07  2.085e-07
B  2.085e-07  2.673e-07
(largest entry is 0.117% of the largest sample covariance entry)
```

The ful residual covariance VaR and the delta normal VaR agree in this case. This is because OLS is the same thing as the joint multivariate Normal on Y and X. The delta normal takes the stocks' joint covariance directly from the data. The regression builds that covariance using the betas with the market variance plus the residual covariance. OLS makes the regressors uncorrelated with the residuals which is why the market part and the residual part just add. The two would stop agreeing as soon as any piece is non-Normal, so for example if errors would be t-distributed or if OLS assumptions were to fail (like market-correlated errors or unstable betas).